Gaussian equivalence and feature learning in neural networks

STATS 200C: High-Dimensional Statistics

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Motivation

- Neural networks (NNs) have achieved remarkable performance over a wide range of tasks. It is widely believed that **feature learning** underlies this success.
- There is empirical evidence[1] suggesting that properties of trained networks are strongly influenced by the early stage of optimization.
- A line of research has shown that NNs trained with one large gradient step can learn nonlinear components of the target function.

[1] Golatkar, Achille, and Soatto, "Time matters in regularizing deep networks: Weight decay and data augmentation affect early learning dynamics, matter little near convergence", 2019

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Background

- In recent years, a large body of work in high-dimensional statistics has been devoted to characterizing empirical risk minimization in the asymptotic regime.
- **Gaussian equivalence**[2] (GE) has emerged as a powerful theoretical framework to study this problem. This presentation will focus on a GE-type result for the feature matrix of trained NNs.

[2] Wen et al., *When does Gaussian equivalence fail and how to fix it: Non-universal behavior of random features with quadratic scaling*, 2025

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Present setting

Our work sits at the intersection of three lines of research:

- Random features and kernel asymptotics[3];
- Feature learning in shallow neural networks[4];
- Spiked or anisotropic covariance models[5].

[3] Cui et al., "Asymptotics of feature learning in two-layer networks after one gradient-step", 2024

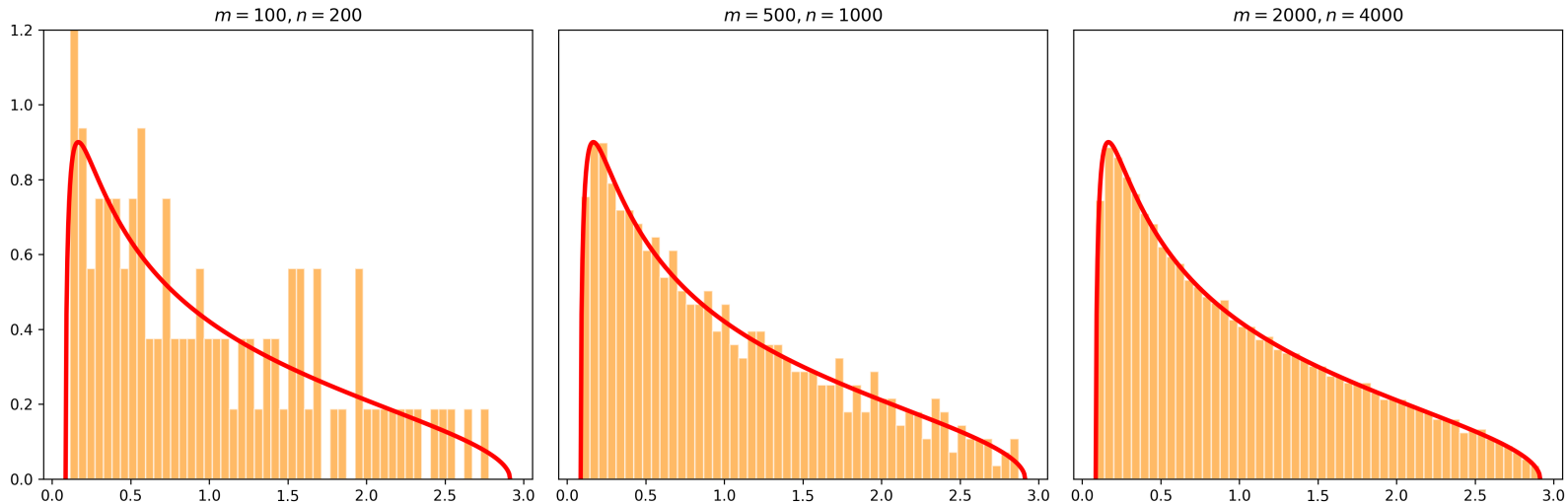
[4] Ba et al., "High-dimensional asymptotics of feature learning: how one gradient step improves the representation", 2022

[5] Wang, Wu, and Fan, "Nonlinear spiked covariance matrices and signal propagation in deep neural networks", 2024

Scaling regime

Our notation follows that of SMM25[6]. We work in the **proportional scaling limit**.

- Let n be the number of training data points, d be the input dimension, and m be the number of hidden neurons. Let $\psi_1, \psi_2 \in \mathbb{R}_{>0}$ be fixed. Consider m and n as functions of d s.t. $\frac{n}{d} \to \psi_1 \in (0, 1)$ and $\frac{m}{d} \to \psi_2$ as $d \to \infty$.

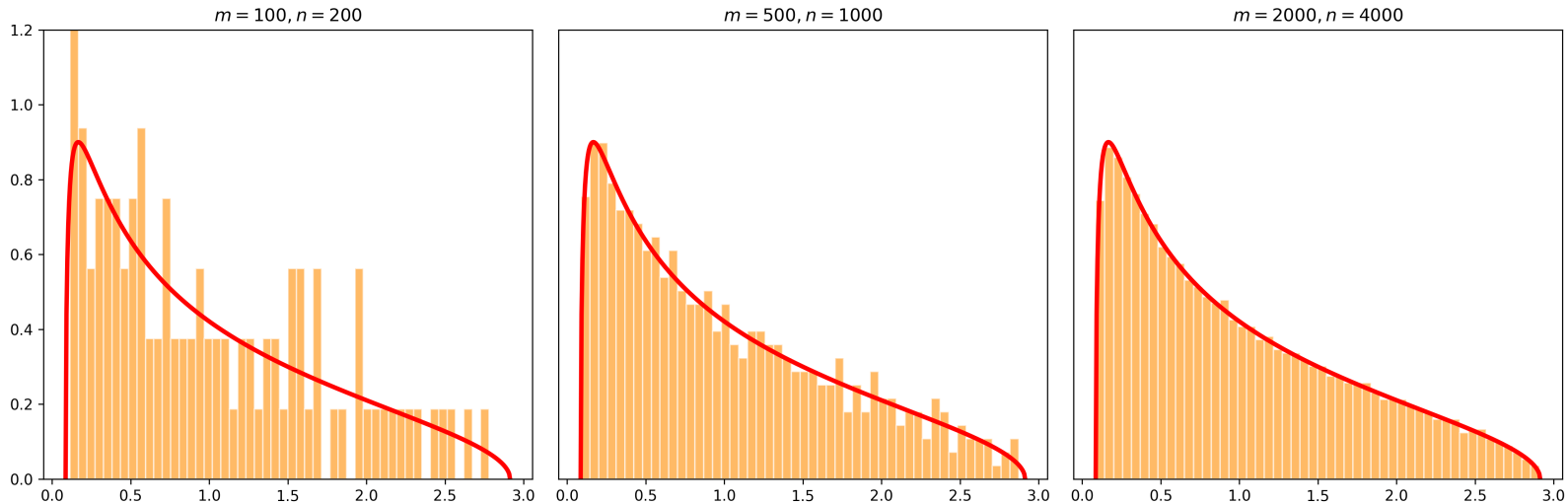


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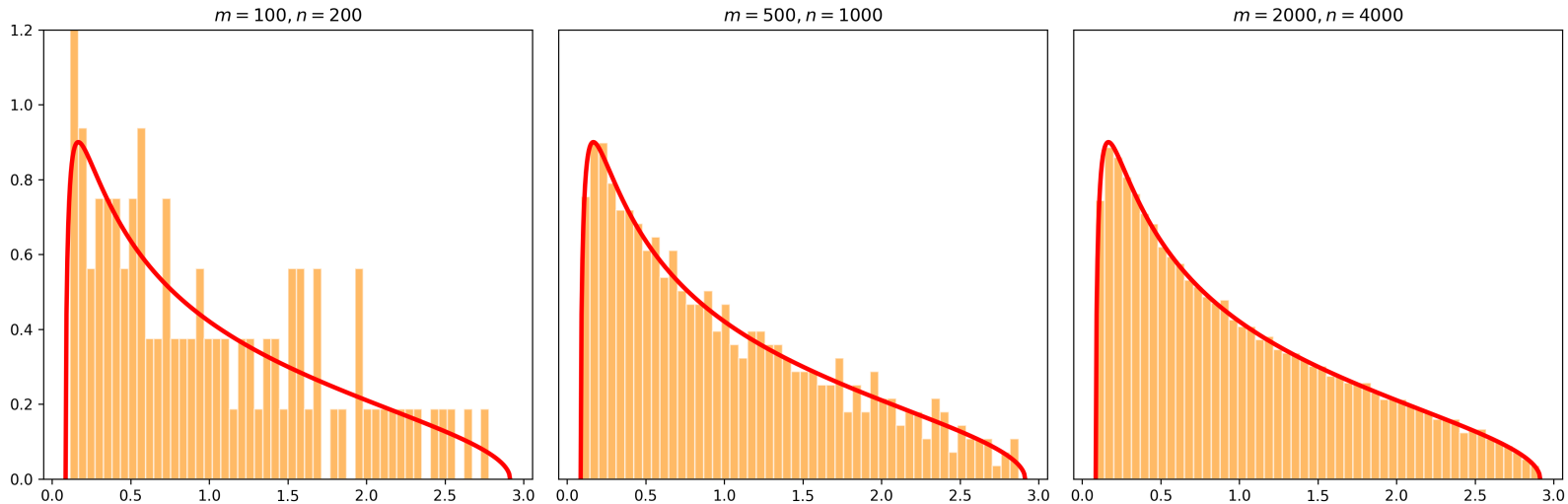


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Present setting

- Our goal is the following: given spiked input data with an anisotropic bulk covariance, we want to characterize how the spikes from the gradient and the input data contribute to the asymptotic test error.
- To that end, we develop and apply a GE property for the features; that result will be the focus of this presentation.

Data model

- **Anisotropic bulk covariance**: Let $\hat{\Sigma} \in \mathbb{R}^{d \times d}$ for which there exists $\alpha \geq 0$ s.t. the k-th eigenvalue satisfies $k^{-\alpha}$ for $k \in [d]$.
- **Spiked data model**: Define $q \in \mathbb{S}^{d-1}$ and $\zeta = n^{\nu}$, where n is the number of data points and $\nu \in [0, 0.5)$ .We assume that $x_1, ..., x_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$, where $\Sigma = \hat{\Sigma} + \zeta^2 q q^\top$.
- **Data decomposition**: We can write X as

$$X = X_B + X_S = X_B + \zeta z q^\top \in \mathbb{R}^{n \times d},$$

where $x_1, ..., x_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \hat{\Sigma})$ and $z \sim \mathcal{N}(\mathbf{0}, I)$.

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Comments: what will $qq^\top$ give, what about $zq^\top$?

Network parameters

Consider a two-layer neural network defined as

$$f(x) = a^\top \sigma(Wx) \in \mathbb{R},$$

where $W = (w_1, ..., w_m)^\top \in \mathbb{R}^{m \times d}$ is the matrix of inner weights, $w_j \in \mathbb{R}^d$ is the weight vector for the j-th hidden neuron, $j \in [m]$, and $a \in \mathbb{R}^m$ is the vector of outer weights.

Training procedure

We follow the training procedure introduced in BES+22[7]:

1. Apply one gradient step to the first layer weights W_0 to get W_1.
2. Solve ridge regression on the updated features $\sigma(W_1 x)$ to obtain the second layer weights $\hat{a}$.

[7] Ba et al., "High-dimensional asymptotics of feature learning: how one gradient step improves the representation", 2022

Regularization

- SMM25[8] show that the first-step gradient G is well-approximated by a low-rank matrix $\hat{G}$.
- We also consider the common regularization technique of ℓ_2 weight decay. Adding the term $\frac{\lambda_{\text{wd}}}{2}\|W_1\|_F^2$ to the loss function yields the modified gradient

$$\hat{G}^{(\lambda)} = \hat{G} + \lambda_{\text{wd}} W_1.$$

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Feature expansion

Given initial weights W_0, gradient approximation $\hat{G}$, and new training data $\tilde{X} = \tilde{X}_B + \tilde{X}_S \in \mathbb{R}^{m \times n}$, we want a decomposition of the form

$$\sigma(\tilde{X} W_1^\top) \approx \underbrace{f(\rho \tilde{X}_B W_0^\top)}_{\text{bulk}} + \underbrace{f(\rho \tilde{X}_S W_0^\top + \eta \tilde{X} \hat{G})}_{\text{spikes}},$$

where $\rho = (1 - \eta \lambda_{\text{wd}})$.

Generalized Hermite polynomials

- The probabilist's Hermite polynomials[9] are popular in statistics because they are orthogonal w.r.t. to the standard Gaussian.
- The anisotropic bulk covariance means we will need Hermite polynomials for generalized Gaussians. Concretely, we assume σ admits the following expansion:

$$\sigma(z) = \sum_{k=0}^{\infty} c_k^t \mathrm{He}_k^t(z), \quad c_k^t = \frac{1}{k!\, t^k} \mathbb{E}_{G\sim\mathcal{N}(0,1)} \left[\sigma(tG)\, \mathrm{He}_k(G)\right].$$

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Feature expansion

Let $\tau_i^2 = w_i^\top \Sigma w_i$. We define

$$F_1 := \begin{bmatrix} \sigma_L^{\tau_1}\left(\tilde{X} w_1^{(1)}\right) \\ \vdots \\ \sigma_L^{\tau_m}\left(\tilde{X} w_m^{(1)}\right) \end{bmatrix}, \qquad F_0 := \begin{bmatrix} \sigma_L^{\tau_1}\left(\rho \tilde{X}_B w_1^{(0)}\right) \\ \vdots \\ \sigma_L^{\tau_m}\left(\rho \tilde{X}_B w_m^{(0)}\right) \end{bmatrix}$$

and

$$S := \rho W_0 \tilde{X}_S^\top + \eta \hat{G} \tilde{X}^\top.$$

Feature expansion

Theorem 1 (Y.–Sonthalia–Montúfar)

Assume certain technical conditions hold. Fix $L \geq 1$. Let $W_0 \in \mathbb{R}^{m \times d}$ be given. Suppose $G \approx \hat{G}$ and

$$W_1 = \rho W_0 + \eta \hat{G}.$$

Then,

$$F_1 = F_0 + \sum_{k=1}^{L} c_k^{\tau} \, \mathbf{1}_n^{\top} \circ S^{\circ k} + \Delta,$$

where Δ is an error term with $\|\Delta\|_2 \leq o(\sqrt{n})$.

Gaussian equivalence

- It has been shown that GE fails under a large gradient step. To circumvent this, MLH+24[10] apply GE to the untrained features F_0.
- Our analysis is similar in spirit, but our setting introduces several difficulties that prevent us from directly adapting their arguments.

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Additional setup

We introduce some more notation.

- For convenience, define $F_0 \in \mathbb{R}^{n \times m}$ columnwise by $F_0 = [g_1, \ldots, g_m]$, where
$$F_{0,j} = g_j := \sigma_L^{\tau}\left(\tilde{X}_B w_j^{(0)}\right) \in \mathbb{R}^n.$$
- For $\lambda > 0$, define the sample-covariance resolvent
$$R_0 := \left(\frac{1}{n} F_0 F_0^{\top} + \lambda I_n\right)^{-1}.$$
- Define
$$\bar{\tau}_j^2 := \rho^2 \left(w_j^{(0)}\right)^{\top} \hat{\Sigma} \left(w_j^{(0)}\right), \quad (c_{>1,L}^{\bar{\tau}_j})^2 := \sum_{k=2}^{L} k! \bar{\tau}_j^{2k} (c_k^{\bar{\tau}_j})^2.$$

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Additional setup

- We use the Lindeberg exchange method[11] to replace the columns of F_0 with a Gaussian surrogate:

$$\tilde{F}_{0,j} = \tilde{g}_j := \rho c_1^{\bar{\tau}_j} \tilde{X}_B w_j^{(0)} + c_{>1,L}^{\bar{\tau}_j} z_j,$$

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Gaussian equivalence

Theorem 2 (Y.–Sonthalia–Montúfar)

Suppose certain technical conditions hold, with $L, \lambda > 0$ fixed. Then, for each admissible vector pair $u, v \in \mathbb{R}^n$ satisfying $\|u\|_2, \|v\|_2 = O(\sqrt{n})$,

$$\frac{1}{n} u^\top R_0 v - \frac{1}{n} u^\top \tilde{R}_0 v \to_P 0.$$

Remarks

Several remarks are in order:

- Part of the notion of admissibility involves working conditional on a certain high-probability event.
- This result also requires control of the (conditional) moments and covariance of the surrogates $\tilde{g}$.
- Fully understanding the test error actually requires a technical strengthening of Theorem 2.

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Proof sketch

1. Use the Lindeberg exchange method to establish a telescoping identity for the bilinear functional.
2. Reduce the problem to controlling rank-one updates by using the Sherman–Morrison formula to define a leave-one-out resolvent identity.
3. Control each increment using Hölder's inequality to show that the telescoping series is bounded.
4. Use the (conditional) Efron–Stein inequality to bound the variance and prove concentration of the bilinear functional.

Generalization error

It turns out that after a lot of algebra, the limiting test error can be written as

$$\mathcal{L}_{\text{test}}(\hat{a}_0) - \mathcal{L}_{\text{test}}(\hat{a}_1) = \mathcal{T}_n + 2\mathcal{S}_n^\top H^{-1}\mathcal{G}_n - \mathcal{S}_n^\top H^{-1}\mathcal{Q}_n H^{-1}\mathcal{S}_n - 2\left(\Xi_n + \mathcal{S}_n^\top H^{-1}\Omega_n\right).$$

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Discussion and limitations

- Our setting does not really "train" the network: taking one gradient step can instead be interpreted as a form of weight initialization that promotes feature learning.
- The sample splitting procedure is natural if one has pretrained models, but the analysis does not translate to the setting where feature learning and ridge regression use the same data.

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Future work

- How can we generalize this to multiple steps? Recent work[12] has established the first theoretical explanation for the heavy-tailed spectra of weight matrices observed in trained NNs.
- Studying feature learning in the multi-step setting will likely require more advanced machinery (e.g., matrix Dyson equations, matrix martingale concentration inequalities).

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